Plant-based therapies for urolithiasis: a systematic review of clinical and preclinical studies.

PMID: 39042342 | Indexed in PubMed

Urolithiasis, the formation of kidney stones, is a common and severe condition. Despite advances in understanding its pathophysiology, affordable treatment options are needed worldwide. Hence, the interest is in herbal medicines as alternative or supplementary therapy for urinary stone disease. This review explores the use of plant extracts and phytochemicals in preventing and treating urolithiasis. Following PRISMA standards, we systematically reviewed the literature on PubMed/Medline, focusing on herbal items evaluated in in vivo models, in vitro studies, and clinical trials related to nephrolithiasis/urolithiasis. We searched English language publications from January 2021 to December 2023. Studies assessing plant extracts and phytochemicals' therapeutic potential in urolithiasis were included. Data extracted included study design, stone type, plant type, part of plant used, solvent type, main findings, and study references. A total of 64 studies were included. Most studies used ethylene glycol to induce hyperoxaluria and nephrolithiasis in rat models. Various extraction methods were used to extract bioactive compounds from different plant parts. Several plants and phytochemicals, including Alhagi maurorum, Aerva lanata, Dolichos biflorus, Cucumis melo, and quercetin, demonstrated potential effectiveness in reducing stone formation, size, and number. Natural substances offer an alternative or supplementary approach to current treatments, potentially reducing pain and improving the quality of life for urolithiasis patients. However, further research is needed to clarify their mechanisms of action and optimize their therapeutic use. The potential of plant-based therapies in treating urolithiasis is promising, and ongoing research is expected to lead to treatment advancements benefiting patients globally.

[Urate urolithiasis: pathogenesis and possibilities of conservative therapy].

PMID: 30575366 | Indexed in PubMed

Urolithiasis is a widespread chronic disease; its prevalence has been steadily increasing over the past 50 years. Urolithiasis accounts for a large proportion urologic diseases, exceeded only by urinary tract infections and diseases of the prostate. Urate urolithiasis refers to a type of urolithiasis, characterized by the formation of kidney stones consisting of uric acid or its salts. In populations of industrialized countries, uric acid is the second or third most frequently occurring stone-forming substance. The article summarizes the data on the global prevalence of both urolithiasis as a whole and urate urolithiasis in particular. The authors provide a detailed overview of the formation of the current concept of the urate urolithiasis pathogenesis and the management of the disease. The main focus is placed on the possibilities and the role of litholytic (stone-dissolving) therapy for urate urolithiasis and the mechanisms of the action of citrate preparations.

Unlocking New Approaches to Urolithiasis Management Via Nutraceuticals.

PMID: 37608670 | Indexed in PubMed

Urolithiasis, commonly known as kidney stones, is characterized by the formation of hard deposits in the urinary tract. These stones can cause severe pain and discomfort, and their management typically involves a combination of medical interventions and lifestyle modifications. According to the literature,

30% and 50% of urolithiasis cases recur. Between 9 and 12% of persons in industrialised countries are predicted to have urolithiasis at some time. Due to the high frequency of stone formation, recurrent nature, and prevalence in adults, it has a significant impact on society, the person, and the health care system. Adopting the best prophylactic measures is crucial in light of these developments to decrease the impact of urolithiasis on individuals and society. In recent years, there has been growing interest in the potential role of nutraceuticals in the management of urolithiasis. Nutraceuticals, such as herbal extracts, vitamins, minerals, and probiotics, have gained recognition for their potential in promoting urinary health and reducing the risk of urolithiasis. These compounds can aid in various ways, including inhibiting crystal formation, enhancing urine pH balance, reducing urinary calcium excretion, and supporting kidney function. Additionally, nutraceuticals can help alleviate symptoms associated with urolithiasis, such as pain and inflammation. While medical interventions remain crucial, incorporating nutraceuticals into a comprehensive management plan can offer a holistic approach to urolithiasis, improving patient outcomes and quality of life. Therefore, nutraceuticals may be a desirable choice for treating and avoiding recurring urolithiasis for patients and medical professionals. Therefore, the present study has focused on nutraceuticals' role in preventing urolithiasis.

Association of nonalcoholic fatty liver disease (NAFLD) with urolithiasis.

PMID: 23363254 | Indexed in PubMed

The NAFLD is related to metabolic disorders and is negatively associated with kidney function. Renal stone disease (urolithiasis) is an increasing form of a common renal disease that is a multifactorial disorder influenced by both intrinsic and extrinsic, mainly environmental factors. The association between the fatty liver and renal calculi, as a specific underlying risk factor, has received no attention, so far. Therefore, in this study, a possible relationship between fatty liver with renal calculi and urolithiasis is investigated. In a cross sectional study, a total of 11245 ultrasonography reports revealing the condition of fatty liver, kidney stones (urolithiasis), or a combination of both of them, were categorized and evaluated statistically. Descriptive statistics determined the number (frequency and percentage) of each condition. The statistical significance of the association between fatty liver and kidney stone, and vice versa, was evaluated using McNemar's test. The Chi Square Test assessed the relationship between genders. Odds ratios and 95% confidence intervals (95% CI) assessed the likelihood of characteristics of urolithiasis for fatty liver patients. We found 8% frequency of urolithiasis among subjects with healthy liver. NAFLD was identified in 30%, while urolithiasis in 11% subjects from all individuals studied. The present study diagnosed urolithiasis in 17% of patients with fatty liver. Its occurrence was more common in men than women. Data revealed more common diagnosis of fatty liver (48%) in patients with urolithiasis, which was also higher in males than females. The higher NAFLD was linked with urolithiasis, indicating a greater chance of their association. Interestingly, the detection frequency of urolithiasis in the patients with NAFLD was also markedly higher (odds ratio: 2.4, 95% CI 2.1-2.7). The NAFLD appears to be an independent variable as a risk factor for stone formation. The present study indicates that the prevalence of urolithiasis is significantly higher in the NAFLD than healthy subjects. This result suggests that NAFLD may be involved in the mechanism of the onset of the urolithiasis. It is suggested that lipid peroxidation, oxidative stress and changes in the urinary constituents in the NAFLD may be considered as a risk factor in the progression of stone formations.

Evaluating the effectiveness of AI-powered UrologiQ's in accurately measuring kidney

stone volume in urolithiasis patients.

PMID: 39527261 | Indexed in PubMed

Kidney stones and urolithiasis are kidney diseases that have a significant impact on health and well-being, and their incidence is increasing annually owing to factors such as age, sex, ethnicity, and geographical location. Accurate identification and volume measurement of kidney stones are critical for determining the appropriate surgical approach, as timely and precise treatment is essential to prevent complications and ensure successful outcomes. Larger stones often require more invasive procedures, and precise volume measurements are essential for effective surgical planning and patient outcomes. This study aimed to compare the ability of artificial intelligence (AI) to detect and measure kidney stone volume via CT-KUB images. CT KUB imaging data were analyzed to determine the effectiveness of AI in identifying the volume of kidney stones. The results were compared with measurements taken by radiologists. Compared with radiologists, the AI had greater accuracy, efficiency, and consistency in measuring kidney stone volume. The AI calculates the volume of kidney stones with an average difference of 80% compared with the volumes calculated by radiologists, highlighting a significant discrepancy that is critical for accurate surgical planning. The results suggest that artificial intelligence (AI) outperforms radiologists' manual calculations in measuring kidney stone volume. By integrating AI with kidney stone detection and treatment, there is potential for greater diagnostic precision and treatment effectiveness, which could ultimately improve patient outcomes.

Herbal medicines in the management of urolithiasis: alternative or complementary?

PMID: 19444769 | Indexed in PubMed

Kidney stone formation or urolithiasis is a complex process that results from a succession of several physicochemical events including supersaturation, nucleation, growth, aggregation, and retention within the kidneys. Epidemiological data have shown that calcium oxalate is the predominant mineral in a majority of kidney stones. Among the treatments used are extracorporeal shock wave lithotripsy (ESWL) and drug treatment. Even improved and besides the high cost that imposes, compelling data now suggest that exposure to shock waves in therapeutic doses may cause acute renal injury, decrease in renal function and an increase in stone recurrence. In addition, persistent residual stone fragments and the possibility of infection after ESWL represent a serious problem in the treatment of stones. Furthermore, in spite of substantial progress in the study of the biological and physical manifestations of kidney stones, there is no satisfactory drug to use in clinical therapy. Data from IN VITRO, IN VIVO and clinical trials reveal that phytotherapeutic agents could be useful as either an alternative or an adjunctive therapy in the management of urolithiasis. The present review therefore critically evaluates the potential usefulness of herbal medicines in the management of urolithiasis.

Oxalate: from the environment to kidney stones.

PMID: 24384768 | Indexed in PubMed

Oxalate urolithiasis (nephrolithiasis) is the most frequent type of kidney stone disease. Epidemiological research has shown that urolithiasis is approximately twice as common in men as in women, but the underlying mechanism of this sex-related prevalence is unclear. Oxalate in the organism partially originate from food (exogenous oxalate) and largely as a metabolic end-product from numerous precursors generated mainly in the liver (endogenous oxalate). Oxalate concentrations in plasma and urine can be modified by various foodstuffs, which can interact in positively or negatively by affecting oxalate absorption, excretion, and/or its metabolic pathways. Oxalate is mostly removed from blood by kidneys and partially via bile and intestinal excretion. In the kidneys, after reaching certain conditions, such as high tubular concentration and damaged integrity of the tubule epithelium, oxalate can precipitate and initiate the formation of stones. Recent studies have indicated the importance of the SoLute Carrier 26 (SLC26) family of membrane transporters for handling oxalate. Two members of this family [Sulfate Anion Transporter 1 (SAT-1; SLC26A1) and Chloride/Formate EXchanger (CFEX; SLC26A6)] may contribute to oxalate transport in the intestine, liver, and kidneys. Malfunction or absence of SAT-1 or CFEX has been associated with hyperoxaluria and urolithiasis. However, numerous questions regarding their roles in oxalate transport in the respective organs and male-prevalent urolithiasis, as well as the role of sex hormones in the expression of these transporters at the level of mRNA and protein, still remain to be answered.

Environmental Conditions as Determinants of Kidney Stone Formation.

PMID: 37913796 | Indexed in PubMed

Urolithiasis is a disease characterized by the presence of stones in the urinary tract, whether in the kidneys, ureters, or bladder. Its origin is multiple, and causes can be cited as hereditary, environmental, dietary, anatomical, metabolic, or infectious factors. A kidney stone is a biomaterial that originates inside the urinary tract, following the principles of crystalline growth, and in most cases, it cannot be eliminated naturally. In this work, 40 calculi from the Don Benito, Badajoz University Hospital are studied and compared with those collected in Madrid to establish differences between both populations with the same pathology and located in very different geographical areas. Analysis by cathodoluminescence offers information on the low crystallinity of the phases and their hydration states, as well as the importance of the bonds with the Ca cation in all of the structures, which, in turn, is related to environmental and social factors of different population groups such as a high intake of proteins, medications, bacterial factors, or possible contamination with greenhouse gases, among other factors.

Kidney Stone Pathophysiology, Evaluation and Management: Core Curriculum 2023.

PMID: 37565942 | Indexed in PubMed

Kidney stone disease, also known as nephrolithiasis or urolithiasis, is a disorder in which urinary solutes precipitate to form aggregates of crystalline material in the urinary space. The incidence of nephrolithiasis has been increasing, and the demographics have been evolving. Once viewed as a limited disease with intermittent exacerbations that are simply managed by urologists, nephrolithiasis is now recognized as a complex condition requiring thorough evaluation and multifaceted care. Kidney stones are frequently manifestations of underlying systemic medical conditions such as the metabolic syndrome, genetic disorders, or endocrinopathies. Analysis of urine chemistries and stone composition provide a window into pathogenesis and direct ancillary studies to uncover underlying diseases. These studies allow providers to devise individualized strategies to limit future stone events. Given its complexity, kidney stone disease is best addressed by a team led by nephrologists and urologists with input from multiple other health professionals including dietitians, endocrinologists, interventional radiologists, and endocrine surgeons. In this installment of AJKD's Core Curriculum in Nephrology, we provide a case-based overview of nephrolithiasis, divided by the individual stone types. The reader will gain a pragmatic understanding of the pathophysiology, evaluation, and management of this condition.

Urolithiasis presenting as right flank pain: a case report.

PMID: 23483000 | Indexed in PubMed

Urolithiasis refers to renal or ureteral calculi referred to in lay terminology as a kidney stone. Utolithiasis is a potential emergency often resulting in acute abdominal, low back, flank or groin pain. Chiropractors may encounter patients when they are in acute pain or after they have recovered from the acute phase and should be knowledgeable about the signs, symptoms, potential complications and appropriate recommendations for management. A 52 year old male with acute right flank pain presented to the emergency department. A ureteric calculus with associated hydronephrosis was identified and he was prescribed pain medications and discharged to pass the stone naturally. One day later, he returned to the emergency department with severe pain and was referred to urology. He was managed with a temporary ureteric stent and antibiotics. This case describes a patient with acute right flank and lower quadrant pain which was diagnosed as an obstructing ureteric calculus. Acute management and preventive strategies in patients with visceral pathology such as renal calculi must be considered in patients with severe back and flank pain as it can progress to hydronephrosis and kidney failure.